Haotian Xue

Ph.D. Candidate in Machine Learning, Georgia Institute of Technology
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Summary

Final-year Ph.D. candidate (expected 2027) seeking full-time research scientist / research engineer roles in **Generative AI**, **Physical AI**, and **Robotics**. Research on diffusion models and their post-training, world models and world action models, and multimodal LLMs; 19 papers (NeurIPS, ICLR, ECCV, TMLR, EMNLP), 570+ citations, h-index 8. Industry research experience at Meta, Adobe, NVIDIA, Microsoft Research, and MIT CSAIL.

Skills

Diffusion models; post-training of diffusion models; world models; world action models; multimodal LLMs; PyTorch and large-scale distributed training.

Education

Georgia Institute of Technology — Atlanta, GA
Ph.D. in Machine Learning (advisor: Yongxin Chen) — Aug 2022 – 2027 (expected)

Georgia Institute of Technology — Atlanta, GA
M.S. in Computer Science — Aug 2022 – Aug 2024

Shanghai Jiao Tong University — Shanghai, China
B.E. in Computer Science, Zhiyuan Honors Program (advisors: Quanshi Zhang, Zhouhan Lin) — Aug 2018 – Jun 2022

Experience

Meta Superintelligence Labs / Reality Labs — May 2026 – present
Research Intern — Mentor: Jiawei Liu
World models and VLM data pipelines.

Adobe Firefly — May 2025 – Nov 2025
Machine Learning Researcher — With Qi Chen, Xun Huang, Eli Shechtman; manager Jinrong Xie
Distillation and motion post-training for video diffusion models (MoGAN, ECCV 2026).

NVIDIA Research (Deep Imagination Research) — May 2024 – Aug 2024
Research Intern — With Jiaojiao Fan, Yao Lu, Jinwei Gu; manager Ming-Yu Liu
Visual grounding and embodied reasoning for vision-language models (Point-It-Out, TMLR 2026).

MIT CSAIL — Sep 2021 – Dec 2022
Visiting Research Intern — With Yunzhu Li, Hsiao-Yu Tung, Josh Tenenbaum
Learning 3D intuitive physics from video (3D-IntPhys, NeurIPS 2023).

Microsoft Research Asia — Sep 2021 – Dec 2021
Research Intern — Mentor: Lei Cui
Large-scale pre-training for document AI.

Preprints

* equal contribution. Full list on Google Scholar.

[1] Lama Moukheiber, **Haotian Xue**, Yongxin Chen. How Long, Not How Close: A Learned Temporal Metric for Planning in Latent World Models. **arXiv 2026**.

[2] Zelin Zhao, Min Shi, Bo Yuan, **Haotian Xue**, Jialuo Li, Lama Moukheiber, Humphrey Shi, Yongxin Chen. WorldBagel: Uncovering the Power of Unified Multimodal Models for Vision-Language-Action-World Modeling. **arXiv 2026**.

[3] **Haotian Xue**, Yipu Chen*, Liqian Ma*, Zelin Zhao, Lama Moukheiber, Yuchen Zhu, Yongxin Chen. ACWM-Phys: Investigating Generalized Physical Interaction in Action-Conditioned Video World Models. **arXiv 2026**.

[4] Lama Moukheiber, Caleb M. Yeung, **Haotian Xue**, Alec Helbling, Zelin Zhao, Yongxin Chen. Beyond a Single Frame: Multi-Frame Spatially Grounded Reasoning Across Volumetric MRI. **arXiv 2026**.

[5] Zeqian Long*, Ozgur Kara*, **Haotian Xue***, Yongxin Chen, James M. Rehg. Immune2V: Image Immunization Against Dual-Stream Image-to-Video Generation. **arXiv 2026**.

[6] Xu Cheng, Xin Wang*, **Haotian Xue***, Zhengyang Liang, Quanshi Zhang. A Hypothesis for the Aesthetic Appreciation in

Neural Networks. **arXiv 2021**.

[7] Hao Zhang, Jiayi Chen, **Haotian Xue**, Quanshi Zhang. Towards a Unified Evaluation of Explanation Methods without Ground Truth. **arXiv 2019**.

Publications

[1] Ozgur Kara, Tunahan C. Ozden, Fatih Horoz, Zeqian Long, **Haotian Xue**, Yongxin Chen, Oguzhan Akcin, Yiting Chen, et al. A Survey on Adversarial Attacks and Defenses for Diffusion Models Across Multiple Modalities. **ECCV 2026 Workshop**.

[2] **Haotian Xue**, Qi Chen, Zhonghao Wang, Xun Huang, Eli Shechtman, Jinrong Xie, Yongxin Chen. MoGAN: Improving Motion Quality in Video Diffusion via Few-Step Motion Adversarial Post-Training. **ECCV 2026**.

[3] Zelin Zhao, Petr Molodyk, **Haotian Xue**, Yongxin Chen. Laplacian Multi-scale Flow Matching for Generative Modeling. **ICLR 2026**.

[4] **Haotian Xue**, Yunhao Ge, Yu Zeng, Zhaoshuo Li, Ming-Yu Liu, Yongxin Chen, Jiaojiao Fan. Point-It-Out: Benchmarking Embodied Reasoning for Vision Language Models in Multi-Stage Visual Grounding. **TMLR 2026**.

[5] Jiaojiao Fan, **Haotian Xue**, Qinsheng Zhang, Yongxin Chen. RefDrop: Controllable Consistency in Image or Video Generation via Reference Feature Guidance. **NeurIPS 2024**.

[6] Atharva Mete, **Haotian Xue**, Albert Wilcox, Yongxin Chen, Animesh Garg. QueST: Self-Supervised Skill Abstractions for Learning Continuous Control. **NeurIPS 2024**.

[7] Yipu Chen*, **Haotian Xue***, Yongxin Chen. Diffusion Policy Attacker: Crafting Adversarial Attacks for Diffusion-based Policies. **NeurIPS 2024**.

[8] **Haotian Xue**, Yongxin Chen. Pixel is a Barrier: Diffusion Models Are More Adversarially Robust Than We Think. **NeurIPS 2024 Workshop on Safe Generative AI**.

[9] **Haotian Xue**, Chumeng Liang*, Xiaoyu Wu*, Yongxin Chen. Toward Effective Protection Against Diffusion-Based Mimicry Through Score Distillation. **ICLR 2024**.

[10] **Haotian Xue**, Alexandre Araujo, Bin Hu, Yongxin Chen. Diffusion-Based Adversarial Sample Generation for Improved Stealthiness and Controllability. **NeurIPS 2023**.

[11] **Haotian Xue**, Antonio Torralba, Joshua B. Tenenbaum, Daniel L. K. Yamins, Yunzhu Li, Hsiao-Yu Tung. 3D-IntPhys: Towards More Generalized 3D-grounded Visual Intuitive Physics under Challenging Scenes. **NeurIPS 2023**.

[12] Shengyuan Hou*, Jushi Kai*, **Haotian Xue***, Bingyu Zhu, Bo Yuan, Longtao Huang, Xinbing Wang, Zhouhan Lin. Syntax-Guided Localized Self-Attention by Constituency Syntactic Distance. **EMNLP 2022 Findings**.

Honors & Awards

NeurIPS Scholar Award	2023, 2024
ICLR Travel Award	2024
NeurIPS Outstanding Reviewer	2023, 2024
NeurIPS Top Reviewer	2022
SJTU Outstanding Graduate	2022
SJTU Zhiyuan Honor Award (top 5%)	2019 – 2022
Singapore Technologies Engineering Scholarship (top 10%)	2019

Academic Service

Reviewer for NeurIPS, ICLR, ICML, CVPR, ECCV, ICCV, AISTATS, AAAI, TPAMI, and TCSVT (since 2021).